

DOUBLE RESPONSES IN DECISION TASKS

— Project Report —
Simulation Based Inference

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Contents

1	Introduction	1
2	Data	2
2.1	Data collection	2
2.2	Source	2
2.3	Previous Usage and Analyses	3
3	Statistical model (simulator)	3
4	Approximator	3
5	Training	4
6	Diagnostics	5
6.1	Convergence	5
6.2	Simulation-based calibration	5
6.3	Parameter recovery	6
7	Inference	6
7.1	Model parameter estimation	6
7.2	Posterior predictive checks	7
8	Discussion & Conclusion	9
8.1	Limitations and potential improvements	9
8.2	Conclusion	9
8.3	Reflection on own learning	10

List of Tables

1	Overview of the dataset.	2
2	Empirical and estimated double response rates.	7
3	Estimated model parameters.	7

List of Figures

1	Example of a simulated trial	1
2	Simulation-based inference pipeline	4
3	Training data diagnostics	5
4	Pairs posterior plot	8
5	ECDF plots	9

1 Introduction

In many cognitive experiments, under time pressure the participants make decisions, and the researchers record two things. First is what they chose and second is how long did it take for them to get to that decision. But, these values are ambiguous in nature, because a slow correct response and a fast correct response come from the same underlying process under different noise levels. This underlying process is what we need to model and that is why models like Evidence Accumulation Models (EAM) exists. So, rather than just looking at the response time and the choice as simple numbers it models the actual decision process.

However, there's a behaviour that gets ignored in almost every analysis of this kind is the double response. A rapid button press immediately after the first one, for the other alternative is usually called the Double response. Most studies treat these double responses as noise and drop them from the analysis. But Evans et al. (2020) argue that the double response rate actually tells something about how the decision process works. If the two responses are inhibiting each other during the accumulation, the losing accumulator gets suppressed as soon as the the winner crosses the threshold, so that is when the double responses should be rare. But, if the accumulators race independently with no inhibition, the loser keeps on accumulating evidence and can still cross the threshold after a while, which produces more double responses. This is why instead of discarding the double responses out, this project actually uses them as a signal.

In this project we are implementing the simplest EAM model called Racing Diffusion Model (RDM), where two accumulators move towards a single threshold with no inhibition. And we have extended the base RDM into the double response decision rule and fit the model using amortized neural posterior estimation with BayesFlow.

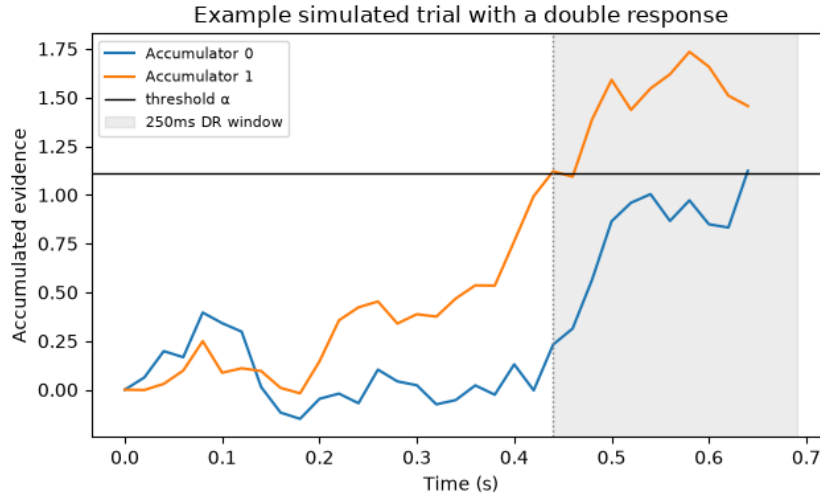


Figure 1: Example of a simulated trial from the double-response RDM ($\nu = [2.24, 0.85]$, $\alpha = 1.11$, $\tau = 0.55$). Accumulator 1 reaches the threshold at $t = 0.44$ s, followed by accumulator 0 at $t = 0.64$ s within the 250 ms double-response window.

We first simulate the double-response RDM using appropriate priors for the drift rates, threshold, and non-decision time. This gives us synthetic datasets where the true parameter values are known. We then train a neural network, referred to as an approximator, to learn the relationship between these simulated datasets and the parameters that generated them. Once trained, the same approximator can be applied to new datasets, including the real experimental data, without having to retrain it each time.

The approximator is first evaluated using simulation-based calibration and parameter recovery. These checks assess whether its posterior estimates are both well calibrated and accurate when applied to data with known parameter values. After this validation, the approximator is applied to the real experimental data, allowing its predictions to be compared with the behaviour observed in the participants.

Since the base RDM that is being used here has no inhibition between the two accumulators, Evans et al. argue that it should not be able to correctly predict the double response rate observed in real data. A mismatch between the predicted and observed double response rates would be consistent with what the theory predicts for the base RDM.

2 Data

2.1 Data collection

Four native Dutch speaker took part in our experiment, and each of them completed 10000 trials, and on each trials, the participants needed to chose whether presented word is a real Dutch word or not. This whole process produced the response time and their choice.

2.2 Source

The experiment was originally conducted by Dutilh et al. (2009). The dataset (*simulated_data.csv*) used in this project was identified as the same dataset by comparing its summary statistics with the one reported by Evans et al. (2020). The basic structure of the dataset is summarised in Table 1.

Table 1: Overview of the dataset.

Quantity	Value
Subjects	4
Trials per subject	10000
Total rows	40000
Columns used	Sub, list_id, Correct, Time1, DoubleResp, Time2

2.3 Previous Usage and Analyses

The original 2009 study used the data to examine how performance on the task changed with practice, focusing on response times and accuracy. Later in the paper by Evans et al. (2020), they used the same dataset to analyse the double responses, which gave us the model that was used in this project.

3 Statistical model (simulator)

We have three parameters that are drift rate, the threshold and the non-decision time and each of them are given proper priors. The drift rate of each accumulator is following Gamma distribution with shape and scale as 5 and 0.5 respectfully. That gives us a mean of 2.5. Similarly for the threshold is also drawn from Gamma distribution with shape and scale as 5 and 0.2 respectively, giving a mean of 1. The non-decision time follows an Exponential distribution with the scale of 0.15. The chosen model assumes that the values of the parameters must be strictly positive, hence all these three parameters are positive as well.

For our project, the starting point for both the accumulators are at zero and the step size is set to 0.02 seconds. At each step, the accumulators increases according to its drift rate, and the step size with some random noise added. If we fix the noise, it keeps the model well defined while allowing the drift rate and the threshold to vary. Both the accumulators continue to move towards the accumulators even after the first one reaches the threshold, as we need the second response to cross the threshold for the double response to occur. The first response is considered as the primary response while the crossing time added with the non-decision time gives us the first response time. As per the second response, if it crosses the threshold within the window of 250 milliseconds, the trial is classified as a double response and the time between the two crossings is considered as the second response time.

4 Approximator

Posterior inference for this project uses BayesFlow. It is a framework built for amortized simulation based inference, that trains a neural network on a simulated data and then later can be reused on any new dataset without retraining.

The summary network is a DeepSet, which treats the trials as an exchangeable set, since the order of the trials does not carry any meaning here. It consists of two equivariant layers, each using small MLPs with two hidden layers of 64 units. Information is combined using mean pooling within and across trials that compresses this into 16 values. The network uses the SiLU activation function, He normal weight initialization, and a dropout rate of 0.05 for regularization.

The inference network is a CouplingFlow, which is a normalizing flow made up of six stacked coupling layers. Each layer applies an affine transformation, with a random permutation between layers and an ActNorm step to help keep training stable. The

transformations inside the coupling layers are handled by MLPs with two hidden layers of 256 units. These use the Mish activation function, residual connections, and a dropout rate of 0.05.

Both networks were left at BayesFlow’s default settings, so no architecture tuning was carried out specifically for the double response RDM.

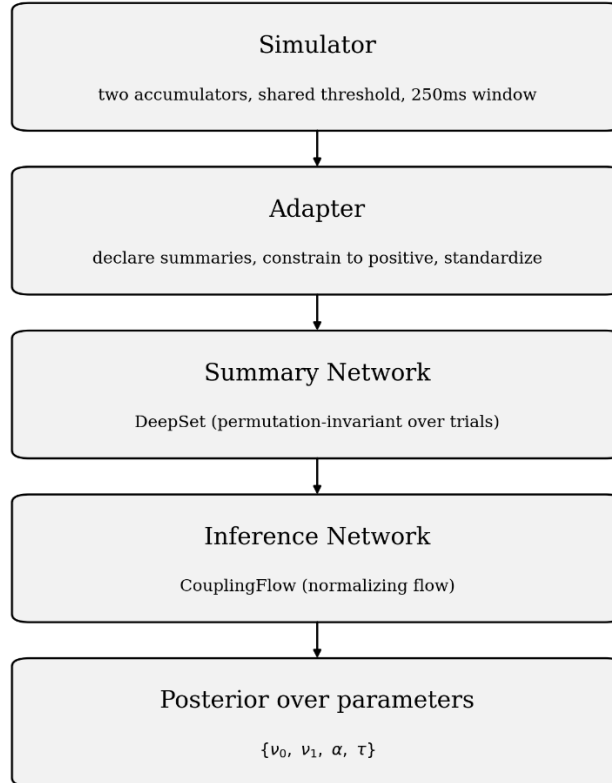


Figure 2: Simplified simulation-based inference pipeline for the double-response RDM.

5 Training

Training was conducted over 100 epochs, number of batches per epoch being 50 over a batch size of 32. We used online regime to ensure that we get new simulated data per epoch. With about $100 \text{ epochs} \times 50 \text{ batches/epoch} \times 32 = \text{effectively } \sim 160000$ training simulations seen over training

```
workflow.fit_online(epochs=100, num_batches_per_epoch=50, batch_size=32)
```


The training took about 31.24 mins and we saved our weights in .keras file format using:

```
workflow.approximator.save("../results/trained_model_v2.keras")
```

The model was trained using the Adam optimizer ($\beta_1=0.9$, $\beta_2=0.999$) with gradient norm clipping (clipnorm=1.5) and a cosine-decay learning rate schedule with linear warmup LR warms up from 5×10^{-5} to a peak of 5×10^{-4} over 250 steps, then cosine-anneals to 0 over the remaining 4,750 steps, matching the full 5,000-step training budget (100 epochs $\times$ 50 batches/epoch).

6 Diagnostics

The diagnostics were conducted on RDM model with (i) convergence checks, (ii) simulation-based calibration (rank histograms and ECDF-difference bands), and (iii) posterior behaviour (contraction and parameter recovery). We included all the results while discussing few (Parameter recovery and ecdf_difference) on a more detailed level.

6.1 Convergence

The model seems to have not been overfit (which was evident as we used online training regime). Moreover, we see a loss of about ~ -2.4 suggesting no underfitting as well as no divergence as loss went down from 5.20 in epoch one to roughly -2.3 to -2.4 by epoch ~ 85 -100.

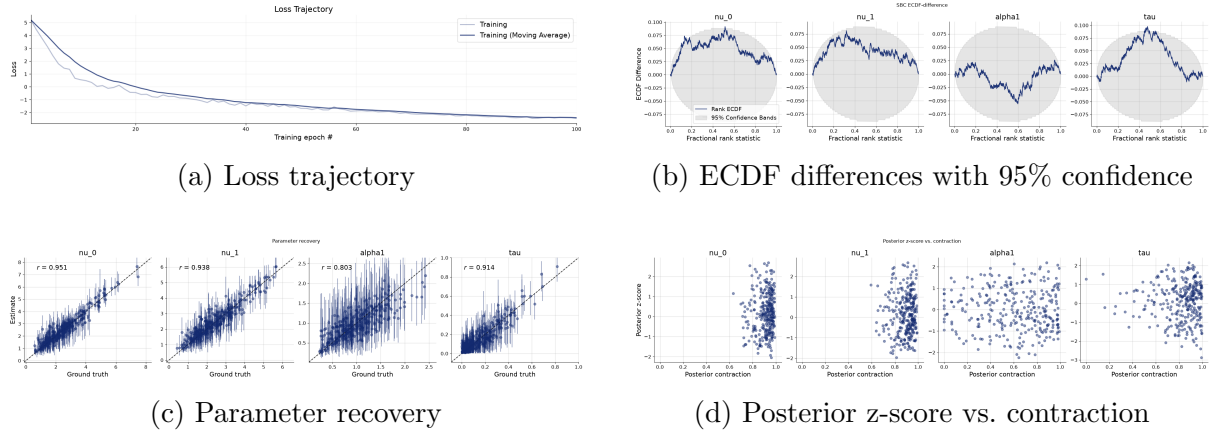


Figure 3: Training data diagnostics: (a) loss trajectories, (b) ECDF differences with 95% confidence bands, (c) parameter recovery, and (d) posterior z-score vs. contraction.

6.2 Simulation-based calibration

We ran SBC and recovery on $N_{test} = 300$ freshly simulated test datasets with $N_{post} = 1,000$ posterior draws each. The approximator can be considered well calibrated, if the rank of each true parameter within its own posterior sample should be uniform across test

dataset ECDF differences (Figure 3b) with 95% confidence ν_0 and τ both sit close to the edge of the 95% confidence band, suggesting a mild systematic drift. Although they show very similar patterns, they arise from two different issues. Looking at ν_0 , we see that the true value regularly landed near the low end of our posterior samples rather than being randomly scattered across the range, which suggests that the model tends to overestimate ν_0 . A possible reason for this is that we are estimating our parameters from the output of reaction time, and reaction time can be thought of as approximately $\frac{\alpha}{\nu_0}$, showing a classical structural identifiability issue. The drift for τ arises from a somewhat different reason: since τ has no independent channel of its own within the RDM, it likely absorbs residual variance left over after ν_0 , ν_1 , and α_1 have been estimated a structural feature of the RDM rather than a separate identifiability problem. This residual-absorption role is consistent with the pairs posterior plots (Figure 4), where τ shows a negative correlation with ν_0 , ν_1 , and α_1 across all four subjects, in contrast to the positive correlation between ν_0 and α_1 that reflects their shared threshold-to-drift identifiability channel.

With α we see a more complex miscalibration pattern where we see non monotonic deviations from the calibration band, meaning it is neither over or under estimating. Since, α does not have a clean independent signal it is not only directly being effected due to the threshold drift ration as discussed with ν_0, ν_1 but also to a certain extent with τ . So, we can conclude it is actually being effected not just by one but 2 different identifiability issues.

6.3 Parameter recovery

Moving forward with parameter recovery (Figure 3c) we see that an excellent recovery with 3 parameters in general where as for alpha we see a sightly lower r values of 0.803 meaning this is also linear with ground truth but not as much as the ν_0, ν_1 and τ . As the true threshold increases, in α we see recovery degrades, posterior estimates show wider credible intervals and greater deviation from the identity line at higher ground-truth α values, consistent with α overall weaker recovery ($r = 0.803$). This is because higher α values fall into a region of parameter space where the $\alpha - \nu_0$ trade-off is more severe: a wider range of (α, ν_0) combinations can produce similar reaction time distributions, leaving the posterior less constrained at the high end of α 's range.

7 Inference

7.1 Model parameter estimation

The model was fit to data from the first experiment of Evans et al. (2020) where 4 native Dutch speakers pressed a button when a letter string, presented at each trial, was either a Dutch word or not. Each participant completed 10,000 trials but analyses were based on a subset comprising 1000 trials per participants. A summary of empirical and estimated double response rates are presented in table 2 below. Double response rates were overpredicted, with relatively narrow confidence intervals and negligible variations

between estimated means and medians, across all four participants. A case in point is the first participant's rate of 2% which was predicted at 16.6%.

Table 2: Empirical and estimated double response rates.

Participant	Empirical		Estimated	
	Mean	Median	Mean(95% CI)	Median (iqr)
1	0.020	0.000	0.166 [0.136, 0.192]	0.165 (0.022)
2	0.034	0.000	0.129 [0.106, 0.153]	0.129 (0.018)
3	0.001	0.000	0.050 [0.031, 0.077]	0.049 (0.016)
4	0.000	0.000	0.067 [0.044, 0.093]	0.067 (0.016)

iqr: interquartile range

The estimated model parameters with corresponding credible intervals are summarized in table 3 below. Overall, the estimated drift rates – ranging from 0.106 to 0.394 – for the incorrect target ($\hat{\nu}_0$) were much lower than those for the correct target ($\hat{\nu}_1$) which ranged from 1.206 to 3.581. The estimated non-decision times ($\hat{\tau}$) were very short with the maximum lasting 0.044 seconds. Comparing the first two participants who had significantly higher double response rates with the third and fourth, the model estimated much lower drift rates for the correct target (1.206 and 2.023 vs 3.581 and 3.085), slightly lower decision boundaries ($\hat{\alpha}_1$) (0.899 and 1.241 vs 1.964 and 1.712), and shorter non-decision time (0.005 and 0.016 seconds vs 0.044 and 0.026 seconds).

Table 3: Estimated model parameters.

Participant	$\hat{\nu}_0$ (95% CI)	$\hat{\nu}_1$ (95% CI)	$\hat{\alpha}_1$ (95% CI)	$\hat{\tau}$ (95% CI)
1	0.106 [0.045, 1.016]	1.206 [0.217, 1.437]	0.899 [0.810, 1.012]	0.005 [0.000, 0.023]
2	0.195 [0.074, 1.785]	2.023 [0.378, 2.293]	1.241 [1.085, 1.398]	0.016 [0.000, 0.058]
3	0.394 [0.111, 3.068]	3.581 [0.875, 4.160]	1.964 [1.537, 2.418]	0.044 [0.001, 0.124]
4	0.275 [0.077, 2.640]	3.085 [0.619, 3.562]	1.712 [1.399, 2.058]	0.026 [0.001, 0.084]

7.2 Posterior predictive checks

Figure 4 illustrates the first participant's mean posterior estimates of the four model parameters with joint distributions displayed off the diagonal and marginal distributions along the diagonal. There was a weak positive correlation between the drift rates towards the incorrect and correct targets. The decision boundary was also positively correlated with the drift rates though the correlation with the rate towards the correct target was stronger. In contrast, there was no correlation between non-decision and the rest of the parameters as the contour plots of their joint distributions were circular.

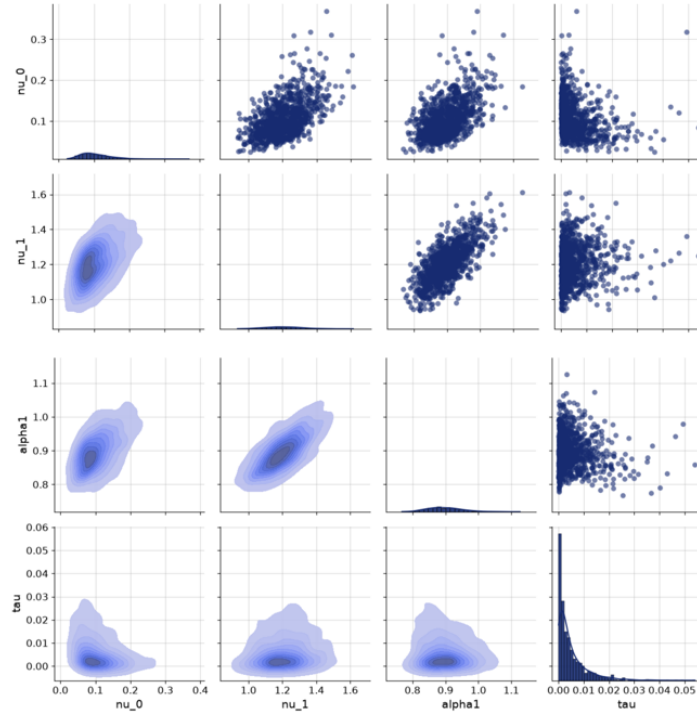


Figure 4: First participant's pairs posterior plot.

The first and third participants' plots of the posterior predictive empirical cumulative distribution (figure 5) were also used to evaluate how well our fitted model replicated the first response time distribution. The ECDFs were calculated for correct and incorrect responses. For subject 1, the model didn't quite fit as the empirical curve for correct responses showed a sharp increase from 0.4 to 1.6 seconds, whereas the posterior predictive median curve increased gradually. This indicated that the model did not fully reproduce the observed response time distribution, notably at shorter response time. However, the posterior predictions captured the general range and ordering of the response times. Conversely, the empirical and posterior predictive curves for subject 3 were closer for both correct and incorrect responses; an indication that the model provided a better representation of this subject's first response distribution.

Importantly, these ECDF checks concern the distribution of the first response time and therefore provide only one aspect of model validation. They do not demonstrate that the model reproduces double response behaviour. This distinction is important because the estimated double response rates were higher than the empirical rates for all four subjects. For example, subject 1 had an empirical double response rate of 0.020, while the posterior predictive mean was 0.166. Similarly, the empirical rates for subjects 2, 3, and 4 were 0.034, 0.001, and 0.000, compared with predicted means of approximately 0.129, 0.050, and 0.067, respectively.

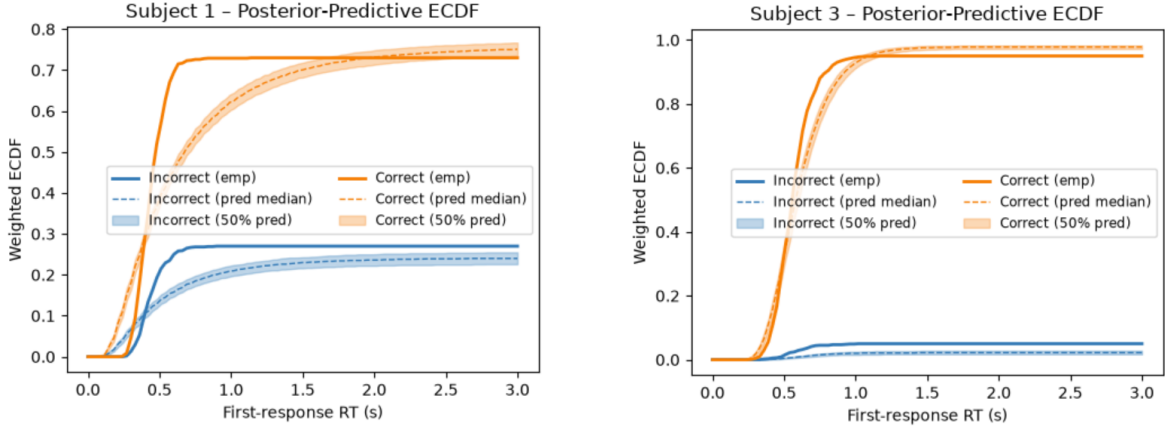


Figure 5: ECDF plots for the first and third participants.

8 Discussion & Conclusion

8.1 Limitations and potential improvements

The main limitation is that the model has no lateral inhibition between accumulator once a response has been triggered, so its condemned to over predict double responses. This isn't really a fault in our pipeline but its RDM nature to behave exactly the way that the theory says an inhibition free model should.

Secondly, we only fit four subjects on a 1000 trial sample whereas originally the subjects completed 10,000 trials as mentioned in the paper. A suggestion would be to test the models on additional cognitive datasets and include more participants to validate better.

Finally, the SBC checks in 6.2 showed ν_0 and τ drifting outside the 95% confidence band, and while we argued this comes from two different mechanisms (a structural $\frac{\alpha}{\nu_0}$ ratio issue for ν_0 , τ absorbing residual variance with no independent channel of its own), we didn't have the compute budget to rerun SBC with a larger test set or posterior draw ratio to rule out that some of this is just sampling noise. A possible improvement would be to introduce an inhibition as discussed by Evans et al. (2020). For example lateral inhibition can reduce the chance of the second accumulator reaching the threshold after the first one. This would reduce the predicted double response rate and give a better representation of the real world scenario. Using our current RDM as the baseline and comparing RDM with lateral inhibition would be the next step.

8.2 Conclusion

In this project, we explored how incorporating double responses into an EAM affects inference and model fit. We chose RDM, the most basic model in Evans et al. (2020)'s family of no inhibition between accumulators. The results show that we were able to recover several parameters reasonably well. The drift rates ν_0 and ν_1 and τ showed

strong parameter recovery, while the threshold α was recovered less accurately because of its identifiability issues. The posterior predictive checks also showed that the model can reproduce important aspects of the first response time distributions, although the quality of the fit varies between participants.

The most important finding would be the double response behaviour. The RDM consistently overpredicted the observed double response rates. Which was consistent with Evans et al. (2020) double response provides an additional constraint that reveals where the model fails to explain the observed behaviour. This supports the main argument of Evans et al. that double responses can be diagnostic of the underlying decision process and can help distinguish between different evidence accumulation models.

Overall, the project shows that good parameter inference does not necessarily mean that the underlying model is adequate. The systematic overprediction of double responses suggests that the RDM is too simple to fully explain the observed data. A model incorporating lateral inhibition would therefore be the next step and could be tested using the same SBI framework.

8.3 Reflection on own learning

One of the main things we learned was that obtaining parameter estimates is not enough to decide whether a model is good. The SBC and parameter recovery checks showed us whether the approximator was able to recover known parameters, while the posterior predictive checks allowed us to see whether the model could reproduce the observed behaviour.

The double response results were especially useful for understanding this difference. Even though the model could recover several parameters reasonably well and reproduce parts of the first response time distribution, it still overpredicted double responses. This showed us that a good inference procedure does not necessarily mean that the underlying model is able to explain the data.

Overall, the project gave us a better understanding of the complete SBI workflow, from defining priors and a simulator, to training the approximator, checking calibration and parameter recovery, and finally evaluating the model using real experimental data.

References

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